Appliance Listing Review

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GitHub repository: https://github.com/emeraldo26/DSO576-Project-

Our vision and usage example

A workforce-development agency must decide which occupations should be prioritized for AI-transition training and reskilling support based on observed AI use, task exposure, employment size, wages, and labor-market outlook.

The program manager filters occupations, compares observed AI exposure with projected employment change, and opens an occupation to see the tasks driving its result. The output is a shortlist for program review—not a prediction that workers will be replaced.

1. Does the data support our vision?

Sources and scope: We will use Anthropic's **job_exposure.csv**, which contains occupation-level observed AI exposure, and **task_penetration.csv**, which contains task-level AI penetration values. We will use Anthropic's bundled O*NET task statements to connect tasks with occupations and O*NET occupation data to provide occupation descriptions. BLS Table 1.2 will provide 2025–35 employment projections, wages, annual openings, education, and training requirements.

The occupation-level files will be joined using **six-digit SOC codes**: occ_code in the Anthropic data, **O*NET-SOC Code** in the O*NET data, and **2025 National Employment Matrix code** in the BLS data. O*NET-SOC suffixes such as **.00** will be removed when necessary. We will also keep only BLS occupational line items and exclude summary categories. For the task-level analysis, we will match the **task** field with the **O*NET Task** field and use the associated O*NET-SOC code to connect each task to an occupation.

Exact column / source	What it contains	Use in our dashboard
Anthropic: job_exposure.csv (occ_code, title, observed_exposure)	Occupation title, SOC occupation code, and observed AI exposure	Create occupation groups, show observed AI exposure in the chart and table, and contribute to the priority score
Anthropic: task_penetration.csv (task, penetration)	Individual occupational tasks and their AI penetration values	Calculate average task penetration and identify the five most exposed tasks
Anthropic's bundled O*NET: onet_task_statements.csv (O*NET-SOC Code, Task)	O*NET task statements connected to occupations	Connect task penetration values to occupation codes
O*NET: onet_occupation_data.csv (O*NET-SOC Code, Title, Description)	Occupation titles and descriptions	Display the description for the selected occupation
BLS Table 1.2 (`2025 National Employment Matrix code`, `Employment, 2025`)	SOC code and employment in thousands	Create the employment-size filter, determine bubble size, and contribute to the priority score
BLS Table 1.2 (`Employment change, percent, 2025–35`)	Projected ten-year employment growth or decline	Create the labor-market-outlook filter, provide the chart's horizontal axis, and contribute to the priority score

BLS Table 1.2 (`Occupational openings, 2025–35 annual average`)	Projected annual openings in thousands	Show continuing demand for workers in the occupation drill-down
BLS Table 1.2 (`Median annual wage, dollars, 2025`)	National median annual wage	Display wage information in the priority table and contribute to the priority score
BLS Table 1.2 (`Typical education needed for entry, Typical on-the-job training needed to attain competency in the occupation`)	Education and training requirements	Provide training and reskilling context in the occupation drill-down

Planned pandas exploration: We will first standardize the SOC codes and task text, then measure how many Anthropic occupations match BLS occupations and how many task records match the O*NET task statements. After checking the match rates, we will create an occupation-level table containing observed AI exposure, average task penetration, employment, wage, projected employment change, and annual openings. We will also rank the tasks within each occupation to identify the five highest penetration values.

What this could support: If the planned joins work, the combined data will support the workforce filters, comparison chart, priority-occupation table, and occupation drill-down shown in our sketch. The priority score will be created after we explore the distributions and decide how the five measures should be weighted.

What is missing: Priority Score formula. We should figure out the appropriate calculation standard for this indicator. Also, the Anthropic data reflects observed Claude use rather than every type of workplace AI use. BLS projections are forecasts and cannot prove that AI caused an occupation to grow or decline. The data also does not include workers' current skills, training-program costs, demographics, state-level demand, or individual job displacement. Therefore, the dashboard should be treated as a decision-support tool, not as proof that AI will replace specific occupations.

2. Dashboard sketch

AI TRANSITION RESKILLING PRIORITIES

dashboard sketch

1. Choose the workforce segment

→ A group includes multiple occupations

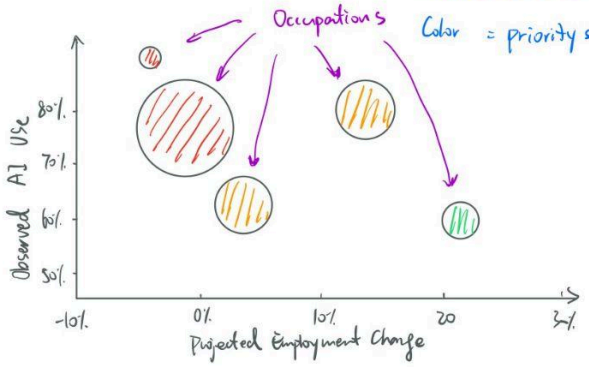
Occupation group: All | Employment size: All | Labor-market outlook: All

2. AI-exposure VS. Labor Outlook

→ Based on the filter

Bubble size = employment size

Color = priority score



Legends:

Priority Score

- High : 70-100
- Medium : 50 - 69
- Lower : below 50

Y : Observed AI exposure

X : Employment Growth

Size : Employment size

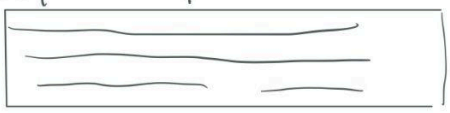
3. Priority Occupations

→ Illustrative values

#	Occupation	Priority Score	Observed AI Exposure	Average Task exposure	Employment	Median Wage	Outlook
1	Customer Service···	84	72%	68%	2.86M	$43k	-5%
2							
3							
4							

4. Occupation Drill-down

- Selected Occupation: XXXX ← Update when the user selects an occupation from the chart or table
- Occupation Description



- Traning and labor-market context

Typical Education	On-the-job Training	Average Annual Openings (2025-2035)
Bachelors	None	xxx k/year

- Top5 Exposed tasks

1 x xx x	95.4%
2 x xxx x	93.7%

How the sketch connects to our data

1. Workforce filters: Users can filter occupations by occupation group, employment size, and labor-market outlook. Occupation groups are based on SOC codes in ***job_exposure.csv***. Employment size comes from *BLS 2025 employment*, while outlook is based on projected employment change from 2025 to 2035 in ***BLS Table 1.2***.

2. Comparison chart: Each bubble represents one occupation. The vertical axis shows observed AI exposure, and the horizontal axis shows projected employment change from ***job_exposure.csv***. Bubble size represents employment, while color indicates the occupation's priority level. The chart updates based on the selected filters.

3. Priority occupations: The table ranks the filtered occupations using a priority score based on observed AI exposure, average task penetration, employment size, median wage, and projected employment change from ***job_exposure.csv, task_penetration.csv, O*NET_task_statements.csv***. The original measures will also be displayed so users can understand why an occupation receives its ranking.

4. Occupation drill-down: When users select an occupation from the chart or table, the dashboard shows its O*NET description, typical education, on-the-job training, and average annual openings. It also displays the five tasks with the highest AI penetration values. Datas are from ***onet_occupation_data.csv, task_penetration.csv, and onet_task_statements.csv***.

Initial scope: The first version will cover the occupations that can be matched across the Anthropic, O*NET, and BLS datasets. The dashboard supports comparison and prioritization, but it does not claim that AI exposure directly causes job loss or automatically recommend a specific training program.